# Bablu Kumar Patel

Roll No.: 25CEM3R06
Master of Technology
Transportation Engineering
National Institute Of Technology, Warangal

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Portfolio
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## Education

**•National Institute Of Technology, Warangal** *2025 – 2027*
*Master of Technology in Transportation Engineering* CGPA: 7.38

**•Rajkiya Engineering College, Azamgarh** *2021 – 2025*
*Bachelor of Technology in Civil Engineering* CGPA: 7.25

**•Ratna Memorial Public School** *2021*
*Board of Senior Secondary Education, Uttar Pradesh* Percentage: 92.6%

## Work Experience

**•Public Works Department (PWD)** *July 2024 – Aug 2024*
*Summer Intern* Gorakhpur, Uttar Pradesh

- Completed a 4-week intensive field exposure program on public infrastructure projects, focusing on road and pavement works.
- Assisted in surveying operations including differential leveling, traversing, and preparation of cross-sections for road alignment.
- Supported quality control of construction materials through soil sample collection, laboratory testing, and pavement layer inspection.

## Personal Projects

**•ML-Based Prediction of Marshall Stability of Waste Modified Asphalt Concrete** *2024 – 2025*
*B.Tech Final Year Project on sustainable materials in asphalt concrete*

- Prepared asphalt mixes with SCBA as filler at 0%, 1%, 3% and 5% replacement levels and reclaimed aggregates as coarse aggregate; varied bitumen content from 4.0% to 8.0%.
- Cast and tested multiple Marshall specimens in accordance with IS and ASTM standards to evaluate Marshall Stability, flow value, air voids and volumetric properties.
- Performed Support Vector Machine (SVM) based modeling on experimental mix design data achieving 99% training accuracy ($R^2$=0.99).
- Demonstrated the potential of SCBA and reclaimed aggregates to improve sustainability of flexible pavements while maintaining acceptable performance.

**•Trip Generation Modeling: Neural Networks vs Linear Regression** *2025*
*Comparative analysis of ML techniques for transportation planning*

- Developed MLP neural network for trip generation modeling achieving $R^2$ of 94.2%, outperforming traditional linear regression by 4.8x.
- Processed 1,800+ household records using feature engineering and data preprocessing techniques.
- Identified earning members and household size as primary predictors through variable importance analysis.

**•Safety Analysis of a Median Opening on a Horizontal Curve at NIT Warangal 2nd Gate** *2025 – 2026*
*Minor Project – Transportation Engineering, NIT Warangal*

- Conducted a field-based traffic safety study on NH-163 at NIT Warangal 2nd Gate, analyzing the combined effect of horizontal curvature, median opening geometry, and mixed traffic behavior on road safety.
- Analyzed Stopping Sight Distance and gap acceptance behavior on NH-163, identifying critical geometric inadequacies where available SSD (25m) fell significantly below required threshold, causing high collision risk.
- Quantified forced merging behavior under mixed traffic – 38.6% of two-wheelers and 25% of cars merged unsafely below critical gap thresholds of 2.5s and 3.5s respectively.
- Proposed geometric and traffic management improvements to reduce conflict points at a complex median opening on a national highway.

**•Pavement Condition Assessment & Overlay Design Using PCI and FWD Analysis** *2026 – 2026*

*M.Tech Course Project – Transportation Engineering, NIT Warangal*

- Performed visual distress survey on an existing pavement stretch and computed Pavement Condition Index (PCI) following IRC:82 guidelines and ASTM D6433 framework to evaluate surface condition and distress severity.
- Analysed Falling Weight Deflectometer (FWD) deflection data in accordance with IRC:115 to back-calculate layer moduli and determine the structural adequacy of the pavement.
- Estimated required bituminous overlay thickness based on deflection-based design approach as per IRC:81 to meet residual service life requirements.

**•Advanced Travel Demand Modelling: Activity-Based Study at NIT Warangal** *2025 – 2026*

*Comparative analysis of mode choice and activity patterns using discrete choice models*

- Conducted an activity-based travel demand study on 144 NITW campus respondents using structured activity diaries, analysing trip chaining behaviour across 285 person-day chains.
- Quantified mode share distribution with walking (56.9%) and cycling (22.9%) accounting for ∼80% of all trips.
- Estimated and compared Nested Logit, Full MNL, and Restricted MNL models for mode choice prediction; Nested Logit was found unsuitable ($\mu > 1$) and Restricted MNL was identified as the best-fit model (AIC: 2043.86, Hit Rate: 62.69%).
- Proposed evidence-based policy recommendations including enhanced pedestrian infrastructure and campus cycling facilities to promote sustainable non-motorised mobility on campus.

## Technical Skills and Interests

**Software**: PTV VISSIM, PTV VISUM, SIDRA INTERSECTION, Civil 3D, IIT PAVE, KGPBACK, STAAD.Pro, AutoCAD (2D Drafting), AIMSUN
**Data Analysis & Modelling**: IBM SPSS, SVM Modelling, Excel, Python (NumPy, Pandas, Matplotlib)
**Laboratory & Materials Testing**: Marshall Stability Testing, Asphalt Mix Design, Aggregate Testing, Concrete Testing, Soil Mechanics (CBR, Compaction, Atterberg Limits)
**Areas of Interest**: Pavement Materials Evaluation and Management, Sustainable Infrastructure, Travel Demand Modeling
**Languages**: English (Fluent), Hindi (Native)

## Positions of Responsibility

**•Media Head,** Innoventure's Club, Rajkiya Engineering College Azamgarh *6 Months*

**•Fest Coordinator,** College Tech Fest, Rajkiya Engineering College Azamgarh

## Certifications & Workshops

**•AutoCAD Training** – Internshala (8-Week Training, Top Performer) *2024*

**•STAAD Pro Training** – Internshala (8-Week Training) *2024*

**•Developing Soft Skills and Personality** – NPTEL

**•Enhancing Soft Skills and Personality** – NPTEL

**•AutoCAD 2D Drafting and Design Workshop** – Learn Delta

## Achievements

- **GATE 2025 Qualified (Civil Engineering)** Successfully qualified in the national-level Graduate Aptitude Test in Engineering.